



KAJIAN SUMBER DAYA ANGIN

Tulungagung, Jawa Timur

Wind Resource Assessment untuk PLTB

Data Angin

ERA5 ECMWF
1980–2025
0.125° Hourly

DEM

DEMNAS BIG
Resolusi 8m
12 Tile

Lokasi

Niyama Beach, Tulungagung
-8.292°S
111.797°E

Turbin Ref.

Vestas V150
4.5 MW
IEC Class III

Hasil Kunci:

Kecepatan Angin Rata-rata @100m: 4.26 m/s

Wind Power Density @100m: 90 W/m²

Capacity Factor: 9.9%

IEC Turbine Class: S (site-specific)

Tanggal Analisis: July 2026 | Analisis berbasis ERA5 + DEMNAS + Statistical Downscaling

Executive Summary — Rangkuman Hasil

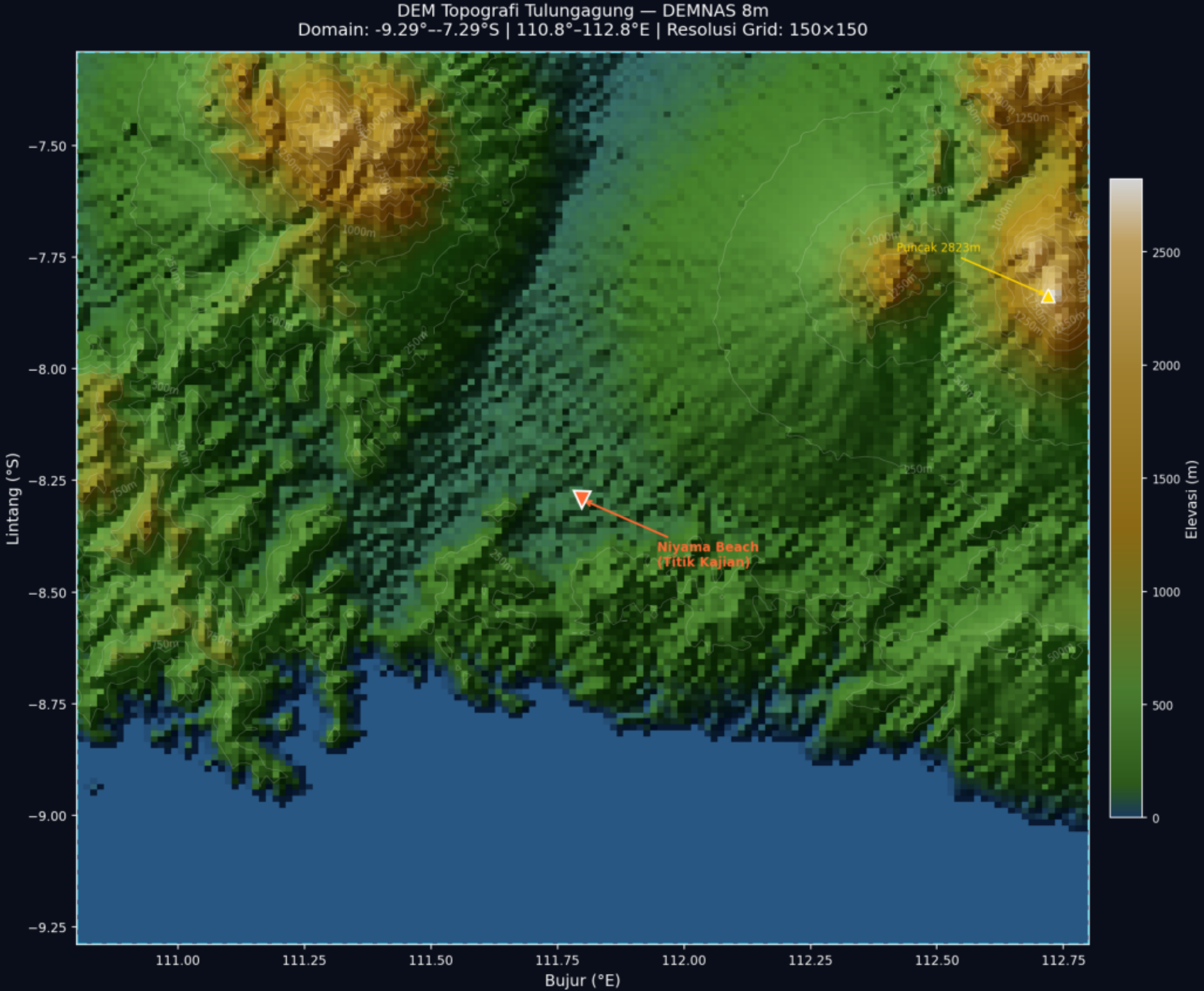
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Parameter	Nilai	Satuan	Keterangan
Kecepatan angin rata-rata (100m)	4.26	m/s	Ekstrapolasi power law $\alpha=0.20$
Wind Power Density (100m)	90.3	W/m²	Formula WPD standar IEC
Capacity Factor estimasi	9.9	%	Vestas V150-4.5MW, Weibull integration
Weibull k / λ (100m)	2.014 / 4.817	— / m/s	MLE fitting scipy.stats
V50-year (100m)	14.1	m/s	Gumbel distribution annual maxima
V_ref IEC (= 1.4 × V50)	19.8	m/s	IEC 61400-1 standar
IEC Wind Turbine Class	Class S (site-specific)	—	Berdasarkan V_ref
AEP layout terbaik (25 turbin)	93.1	GWh/tahun	Layout C_ridge
Wake loss layout terbaik	4.9	%	Jensen/Park model
Tren kecepatan angin	-0.016	m/s/dekade	Linear regression ERA5
Rekomendasi	Kurang layak untuk PLTB skala besar	—	Berdasarkan IEC Class III threshold

REKOMENDASI: Kurang layak untuk PLTB skala besar

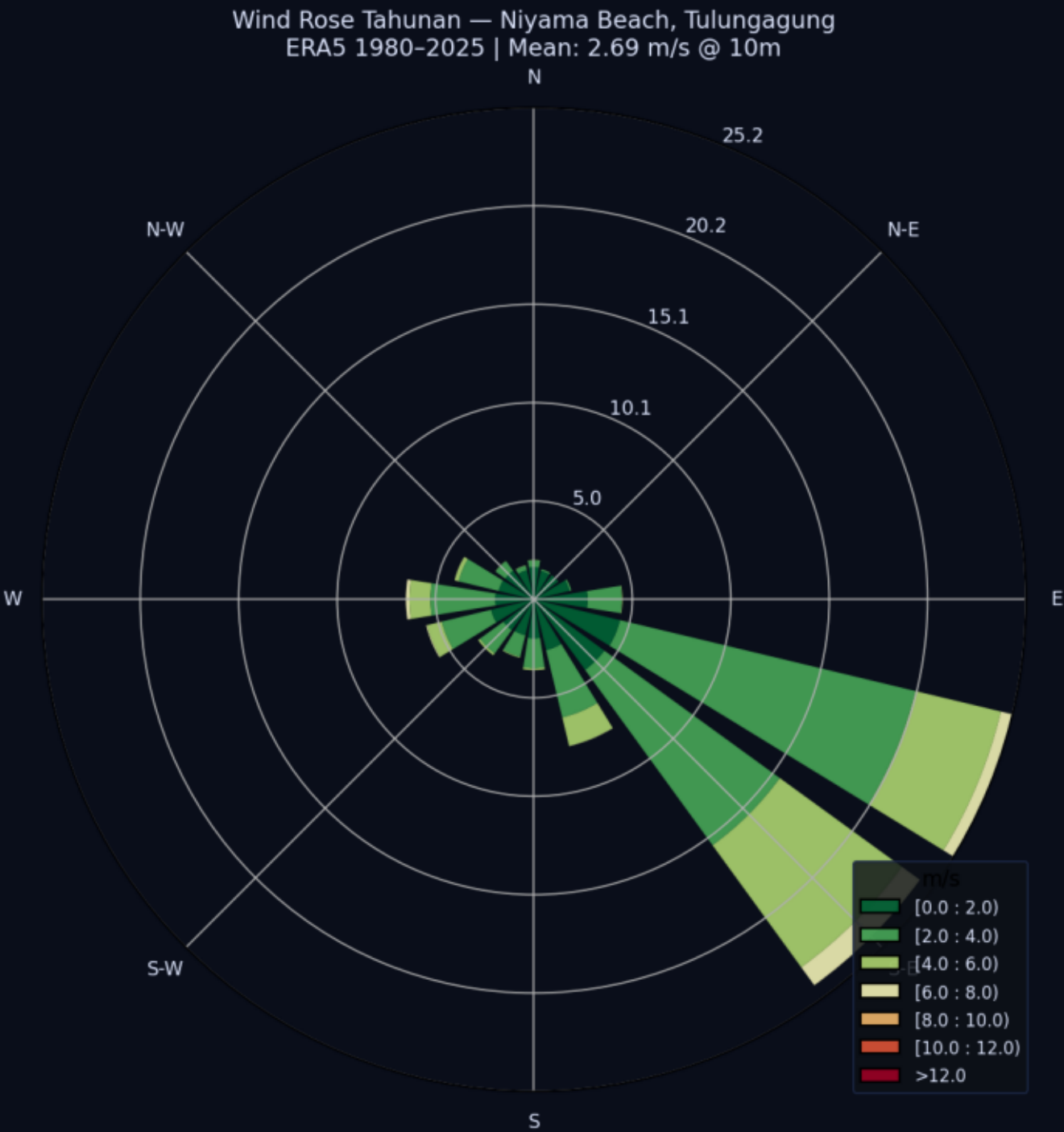
Peta Topografi DEMNAS 8m — Domain Kajian

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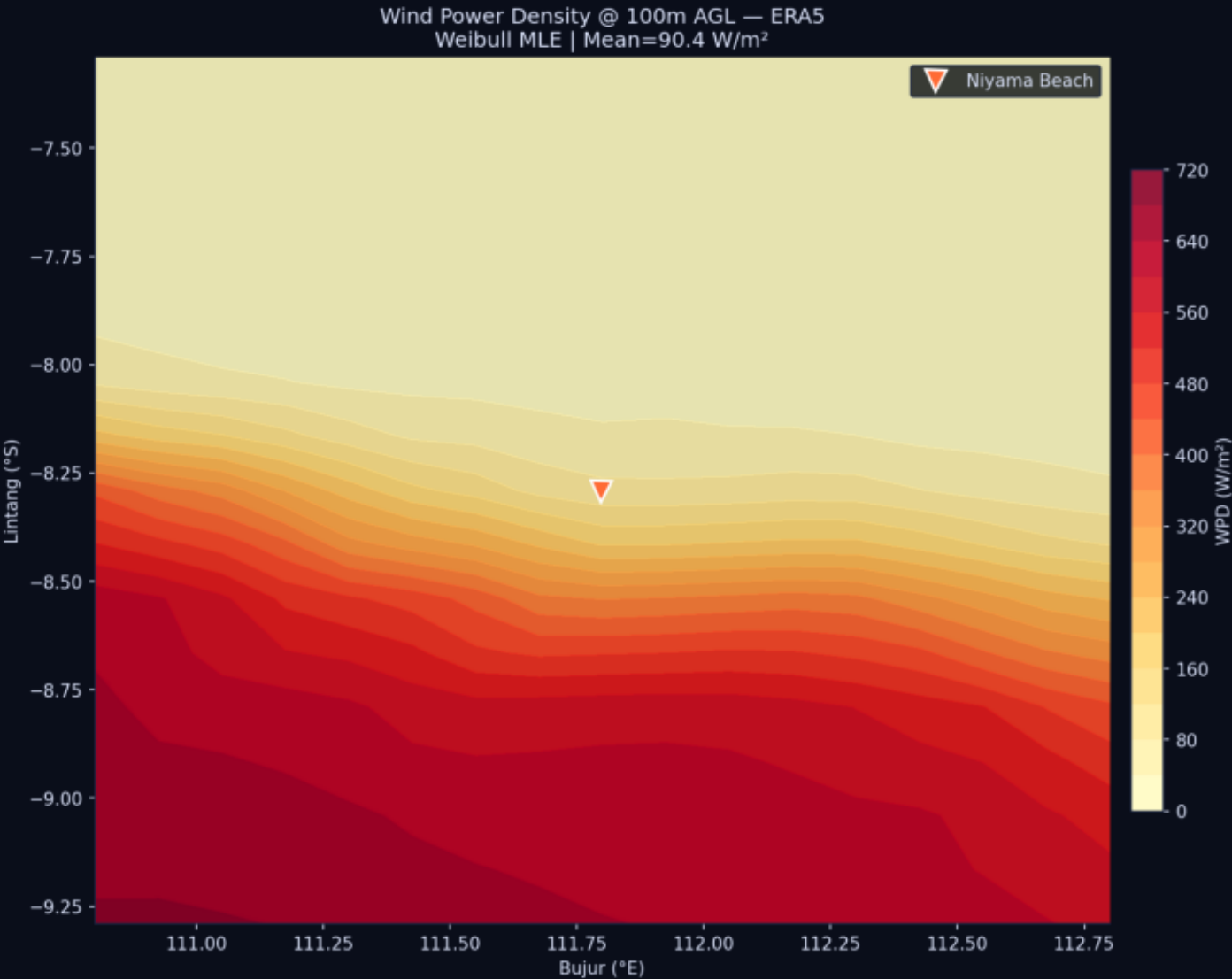
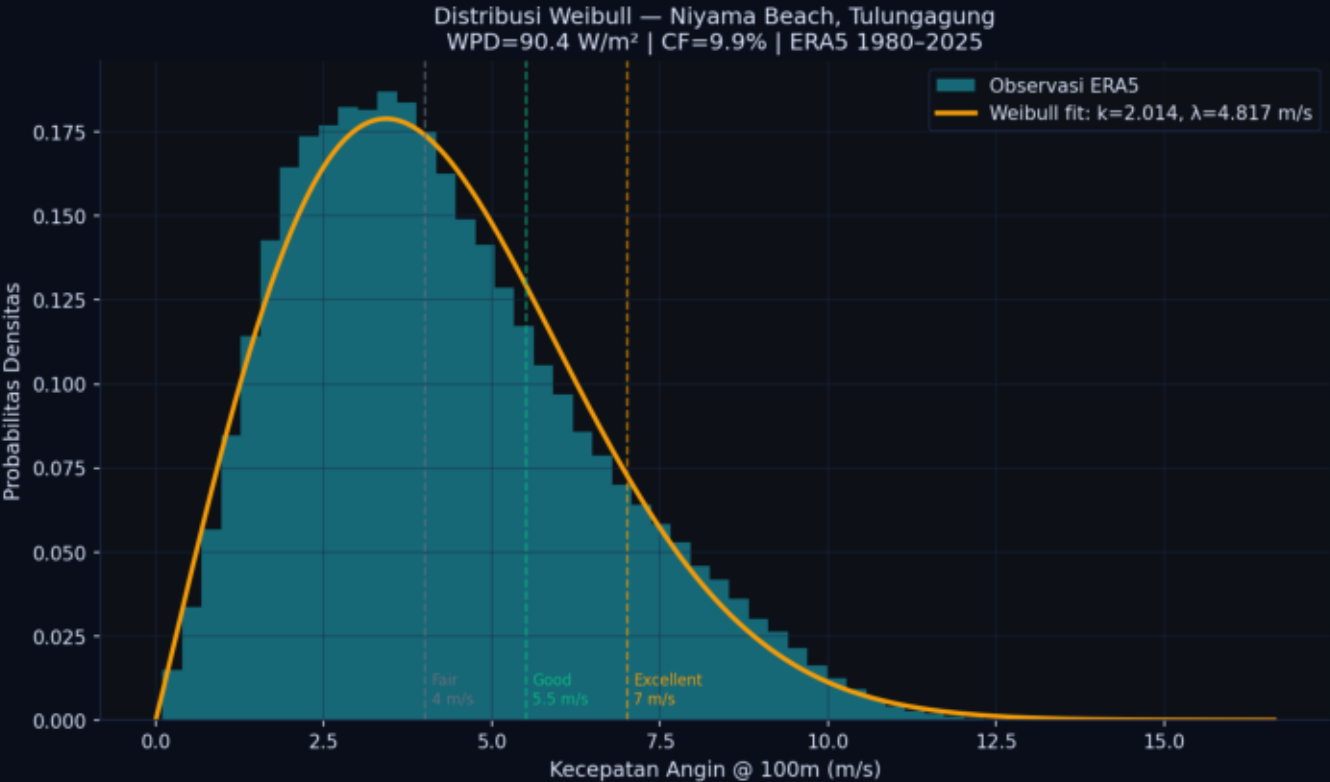
Iklm Angin ERA5 1980-2025

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Distribusi Weibull & Peta Wind Power Density

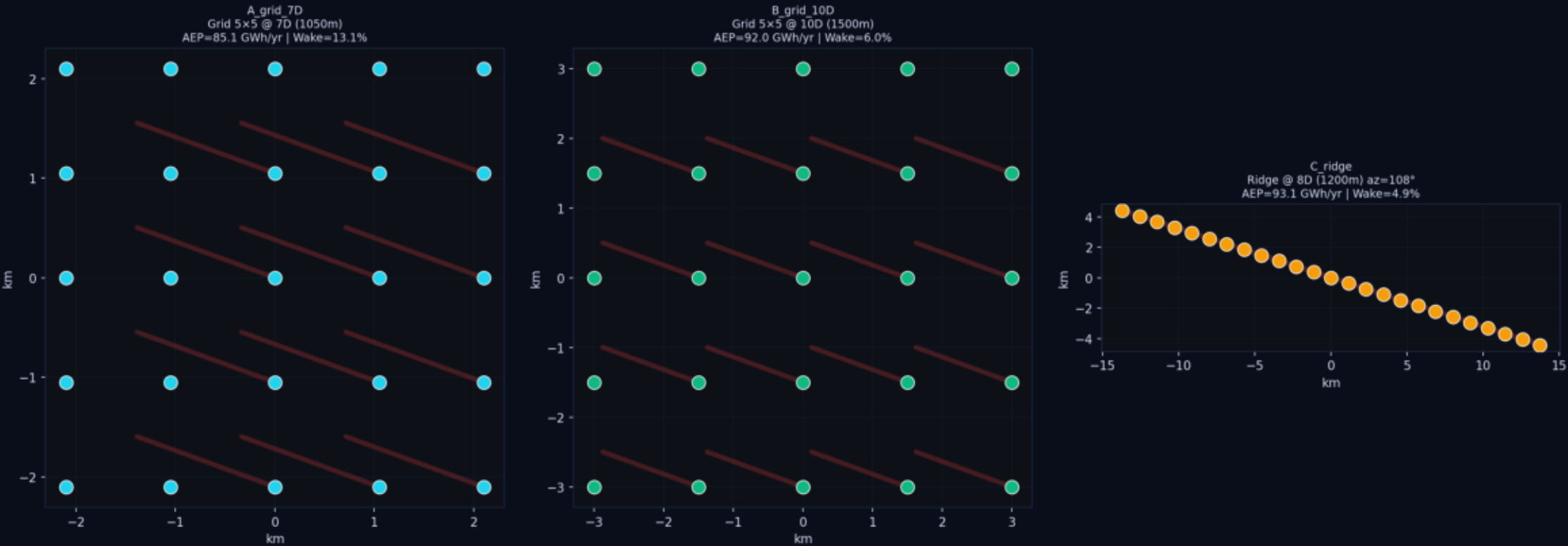
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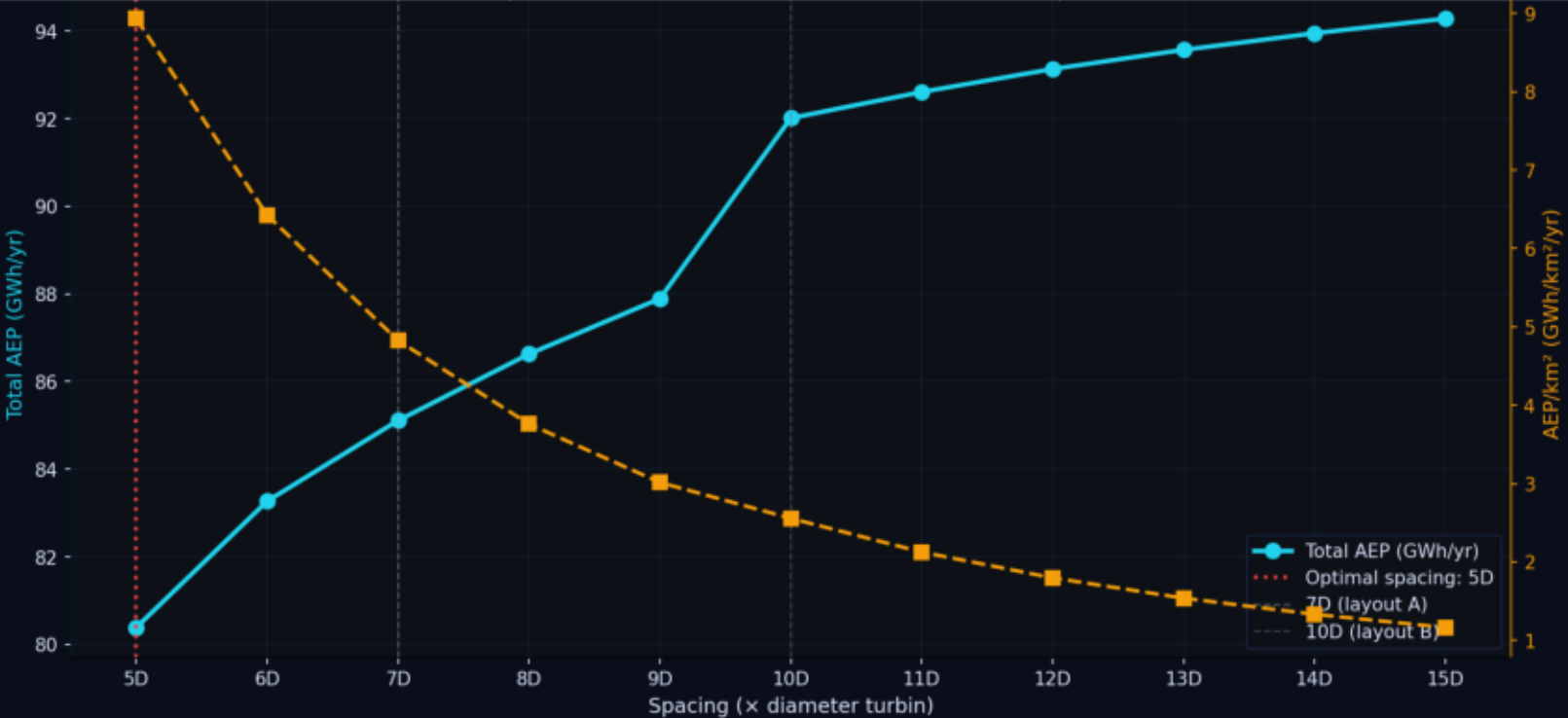
Analisis Wake Effect — Layout Turbin

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Perbandingan Layout Kandidat Wind Farm — Jensen Wake Model

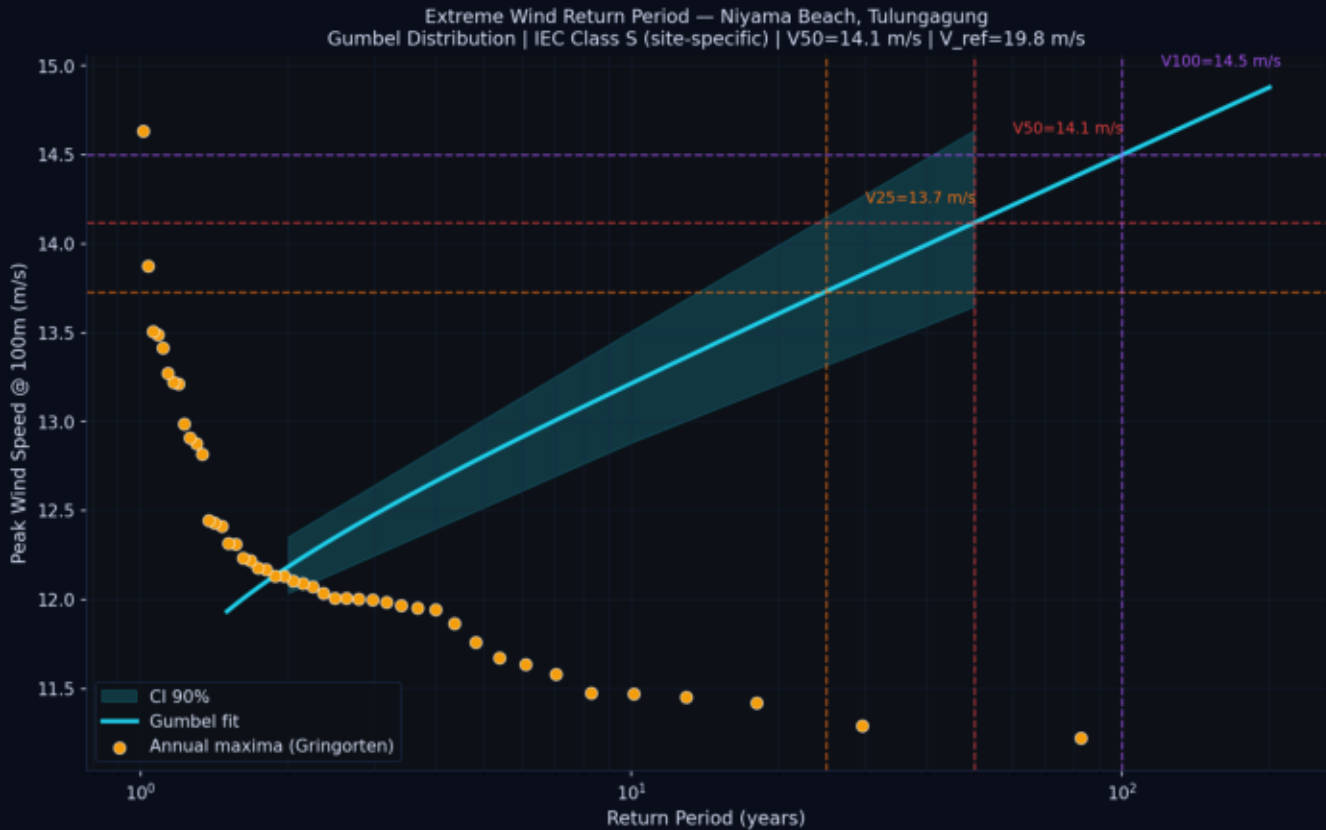


Layout	AEP (GWh/yr)	Wake Loss (%)	Area (km²)	Efisiensi (%)
A_grid_7D	85.11	13.09	17.64	86.91
B_grid_10D	92.01	6.04	36.0	93.96
C_ridge	93.09	4.94	241.6	95.06



Extreme Wind Analysis — IEC Class S (site-specific)

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Return Period (yr)	1	5	10	25	50	100
Peak Wind Speed (m/s)	-inf	12.8	13.21	13.73	14.12	14.5
Design Wind Speed (m/s)	-inf	17.92	18.49	19.22	19.77	20.3



► 1. Data Angin

ERA5 ECMWF reanalysis 1980–2025, resolusi 0.125°, 1-jam.

Variabel: u10, v10 (komponen zonal dan meridional 10m AGL).

► 2. DEM

DEMNAS Badan Informasi Geospasial (BIG) Indonesia, resolusi 8m.

12 tile mencakup domain Tulungagung dan sekitarnya.

► 3. Ekstrapolasi Ketinggian

Power Law: $V(z) = V_{10} \times (z/10)^{\alpha}$

$\alpha = 0.20$ untuk onshore (kondisi netral, terrain terbuka).

► 4. Downscaling Spasial

Jackson-Hunt topographic speedup correction dari ERA5 ke grid 3km.

$speedup = 1 + (\Delta h/L) \times f_{shape} \times exposure_index$.

Roughness correction: $\ln(z/z_o_lokal) / \ln(z/z_o_ERA5)$.

► 5. Statistik Distribusi

Weibull 2-parameter (Maximum Likelihood Estimation, scipy.stats).

$WPD = 0.5 \times \rho \times \lambda^3 \times \Gamma(1+3/k)$. Capacity Factor dari integrasi Weibull x power curve.

► 6. Wake Modeling

Jensen/Park top-hat model.

$U_{wake} = U_{\infty} \times (1 - (1 - \sqrt{1 - C_t}) \times (r_o / (r_o + kx))^2)$. $k = 0.05$ (onshore).

► 7. Extreme Wind

Gumbel GEV fitted ke annual maxima (45 tahun).

Return periods $T=1,5,10,25,50,100$ tahun. CI 90% via bootstrap $N=500$.

► 8. IEC Class

$V_{ref} = 1.4 \times V_{50}$. Class I: ≥ 50 m/s, II: ≥ 42.5 , III: ≥ 37.5 .

► 9. Keterbatasan

Resolusi ERA5 0.125° tidak menangkap variabilitas sub-grid.

Disarankan pengukuran anemometer 2 tahun minimum untuk validasi.

Referensi Ilmiah & Standar Teknis

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Laporan ini dibuat secara otomatis oleh pipeline Wind Resource Assessment Tulungagung.